10-1. The Potential Outcome Framework

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Potential Outcome Framework* (Prof. Sanghack Lee, GSDS SNU; adapted from material by Fan Li and Matthew Blackwell)

The central question is how causal effects are defined and identified when causation is formulated as a comparison of potential outcomes, a missing-data problem, rather than through structural equations and graphs.

These notes consolidate the slides, work the "God's eye" example numerically, prove identification of the ATE under randomisation and under ignorability, and relate the assumptions to the graphical framework of Lectures 2–5. Physical PDF page numbers are listed in Section C.

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1 Frameworks for causal inference

Purpose. Place the potential outcome (PO) framework next to the causal-diagram framework.

A framework for causal inference must be rich enough to formalise basic intuitions about cause and effect. Two are in common use:

- the **potential outcome framework**, also called the counterfactual framework or Neyman–Rubin causal model (Neyman 1923; Rubin 1974);
- the **causal diagram framework** (Pearl 2009).

Mathematically the two are connected (Richardson and Robins 2013; see the SWIG lecture).

The PO framework is arguably the most widely used across medicine, health care, policy and the social sciences. Causal inference is viewed as a *missing-data problem*, with explicit modelling of the *assignment mechanism* as the process that reveals the observed data. It is rooted in Fisher's (1918, 1925) and Neyman's (1923) work on randomised experiments, extended by Rubin (1974, 1978) and many others to non-randomised studies.

2 Basic concepts

Purpose. Define units, treatments, potential outcomes and causal effects, and state the fundamental problem.

- **Unit:** the person, place or thing on which a treatment operates, at a particular time. (The same person at two different times constitutes two units.)
- **Treatment:** an intervention whose effect on some measurement of the units the investigator wishes to assess relative to no intervention (control).
- **Potential outcomes:** the values of a unit's measurement after application of the active treatment, $Y(1)$, and after non-application (control), $Y(0)$. Synonymously, *counterfactuals*.
- **Causal effect:** for each unit, a comparison of $Y(1)$ and $Y(0)$, e.g. $Y_i(1) - Y_i(0)$.

2.1 The fundamental problem of causal inference

At most one potential outcome is observed for each unit; the other is missing, i.e. counterfactual (Holland 1986).

Unit	$Y(x{=}0)$	$Y(x{=}1)$	X	Y
1	?	1	1	1
2	0	?	0	0
3	?	0	1	0
4	?	1	1	1
5	1	?	0	1

Neyman's urn model. Picture each unit as a ticket carrying both Y_1 and Y_0, placed in an urn. Assignment draws tickets and, for each, reveals only one half: Y_1 for treated units, Y_0 for controls.

2.2 God's eye view: individual versus average effects

Outcome Y: death (1) or survival (0). With both potential outcomes visible (adapted from Hernán and Robins, *What If*, Table 1.1):

	$Y(0)$	$Y(1)$	effect		$Y(0)$	$Y(1)$	effect
Rheia	1	0	-1	Hestia	1	0	-1
Zeus	0	0	0	Poseidon	1	0	-1
Kronos	0	1	$+1$	Hera	1	0	-1
Demeter	0	0	0	Artemis	1	1	0
Hades	0	0	0	Apollo	0	1	$+1$
Risk: $\mathbb{E}[Y(0)] = 0.5$, $\mathbb{E}[Y(1)] = 0.3$							

The average causal effect is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = 0.3 - 0.5 = -0.2$: treatment reduces risk. But individual effects vary: 4 helped (-1), 2 harmed ($+1$), 4 unaffected. We can never observe both columns for the same person.

2.3 What the researcher actually sees

(Adapted from *What If*, Table 1.2.)

	$Y(0)$	$Y(1)$	X	Y^{obs}		$Y(0)$	$Y(1)$	X	Y^{obs}
Rheia	?	0	1	0	Hestia	?	0	1	0
Zeus	0	?	0	0	Poseidon	1	?	0	1
Kronos	?	1	1	1	Hera	?	0	1	0
Demeter	0	?	0	0	Artemis	1	?	0	1
Hades	0	?	0	0	Apollo	?	1	1	1

The "?" entries are counterfactual and forever missing. By consistency $Y^{\text{obs}} = Y(X)$. The naive comparison gives $\bar{Y} = 2/5 = 0.4$ among the treated and $2/5 = 0.4$ among controls, a difference of $0 \neq -0.2$. Why? Treatment was not randomly assigned: sicker patients (those with $Y(0) = 1$) chose treatment more often (confounding). This is why we need exchangeability.

3 Assumptions linking potential and observed outcomes

Purpose. State SUTVA and consistency, and illustrate their failures.

3.1 SUTVA

Assumption 1 (Stable Unit Treatment Value Assumption; Rubin 1980). SUTVA comprises (1) *no interference* and (2) *no different versions* of a treatment (e.g. the same surgery performed by different surgeons). It connects the observed treatment X with the intervention of interest x:

$$X_i = 1 \Rightarrow Y_i = Y_i(1), \qquad X_i = 0 \Rightarrow Y_i = Y_i(0),$$

equivalently

$$Y_i = X_i Y_i(1) + (1 - X_i) Y_i(0).$$

Violation by interference: $Y_i(x)$ depends on the treatments received by other units, e.g. vaccination, advertising, infectious diseases, social networks, agricultural plots. Then other assumptions are needed (Rosenbaum 2007; Hudgens and Halloran 2008), and spillover effects are often themselves of interest.

On the history of SUTVA (VanderWeele and Hernán, "Causal inference under multiple versions of treatment"). Rubin (1980) pointed out that the notation $Y_j(a)$ presupposes (i) that j's outcome under a does not depend on the treatments of other individuals and (ii) that there are no multiple versions of a leading to different outcomes. The first is "no interference" (attributed to Cox 1958), the second "no versions of treatment" (attributed to Neyman 1935). Also included in SUTVA is *consistency* (Robins 1986): $Y_j(a) = Y_j$ when $A_j = a$, tying counterfactual data to observed data. These assumptions matter in both experimental and observational studies; although usually stated only for observational research, they also govern the interpretation of randomised trials.

3.2 Consistency and well-defined interventions

Consistency: if $X_i = x$ then $Y_i = Y_i(x)$. It sounds obvious, but requires that "$X = x$" means the same thing for everyone. Example: the effect of "losing 20 kg" on heart-attack risk. Losing 20 kg through diet and exercise ($Y(x)$ = survived) and losing it through starvation or illness ($Y(x)$ = died) are the same nominal treatment with different outcomes, so $Y(x)$ is not well-defined. Fix: specify the intervention precisely ("20 kg loss via a supervised diet and exercise programme"). Vague treatments give vague causal questions and meaningless answers (*What If*, Fine Point 3.4).

4 Causal estimands

Purpose. List the quantities typically targeted.

Differences.

$$\begin{aligned} \text{ATE:} \quad & \tau = \mathbb{E}[Y(1) - Y(0)], \\ \text{CATE:} \quad & \tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z], \\ \text{ATT:} \quad & \text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid X = 1]. \end{aligned}$$

Ratios. Individual ratio $r_i = Y_i(1)/Y_i(0)$; population ratio $r = \mathbb{E}[Y(1)]/\mathbb{E}[Y(0)]$; relative effect (lift) $l = \mathbb{E}[Y(1) - Y(0)]/\mathbb{E}[Y(0)]$. Note that the population ratio is not an average of individual ratios unless the effect is homogeneous. Continuous outcomes (income, test scores) dominate in the social sciences; binary outcomes (mortality, readmission) are more common in medicine and health.

5 Randomisation and observational studies

Purpose. Show why randomisation identifies causal effects and which assumptions replace it in observational data.

5.1 Unconfounded assignment

An assignment mechanism is *unconfounded* if it does not depend on the potential outcomes, only on pre-treatment covariates:

$$p(X = 1 \mid Z, Y(0), Y(1)) = p(X = 1 \mid Z).$$

$p(X = 1 \mid Z)$ is the *propensity score* (Rosenbaum and Rubin 1983). In randomised experiments the mechanism is unconfounded, and moreover known and controlled by the investigator.

5.2 The role of randomisation

Randomisation balances observed covariates ($X \perp\!\!\!\perp Z$), unobserved covariates ($X \perp\!\!\!\perp U$) and potential outcomes, i.e. it guarantees ignorability

$$X \perp\!\!\!\perp (Y(1), Y(0)).$$

("Covariates", "pre-treatment variables" and "baseline characteristics" are used interchangeably for Z.) Under randomisation causal effects are identified:

$$P(Y(x)) = P(Y(x) \mid X = x) = P(Y \mid X = x), \qquad x \in \{0, 1\},$$

the first equality by ignorability and the second by consistency. Association then implies causation (within the PO framework and its assumptions):

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid X = 1] - \mathbb{E}[Y(0) \mid X = 0] = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0].$$

As graphs. In a randomised experiment nothing points into X (e.g. $X \to Y \leftarrow Z$). A single-world intervention graph (SWIG) splits the treatment node into the observed X and the intervened value x, making potential outcomes explicit: $X \mid x \to Y(x) \leftarrow Z$. Since X and $Y(x)$ are d-separated, $Y(x) \perp\!\!\!\perp X$.

5.3 Observational studies

The most common assumptions:

1. **No unmeasured confounding:** $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$. Also called unconfoundedness, ignorability, selection on observables, no omitted variables, exogeneity, conditional exchangeability. Conditional on covariates, X_i is effectively randomly assigned.
2. **Positivity (overlap):** $0 < P(X_i = 1 \mid Z_i) < 1$; both treatment and control are possible at every value of Z_i.

We take Z as given for now; choosing it is exactly the adjustment-criterion problem. These assumptions can be wrong. Graphically, $Z \to X$, $Z \to Y$, $X \to Y$; in the SWIG, $Z \to X$, $Z \to Y(x)$, $x \to Y(x)$, and $Y(x) \perp\!\!\!\perp X \mid Z$ is read off by d-separation.

5.4 Identification of the ATE

Under positivity and no unmeasured confounding,

$$\begin{aligned}
\tau &= \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_Z\{\mathbb{E}[Y(1) - Y(0) \mid Z]\} = \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid Z] - \mathbb{E}[Y(0) \mid Z]\} \\
&= \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid X = 1, Z] - \mathbb{E}[Y(0) \mid X = 0, Z]\} && \text{(ignorability)} \\
&= \mathbb{E}_Z\{\mathbb{E}[Y \mid X = 1, Z] - \mathbb{E}[Y \mid X = 0, Z]\} && \text{(consistency).}
\end{aligned}$$

Positivity ensures that the conditional expectations in the last line are defined for every z. Define the treated and control conditional expectation functions (CEFs)

$$\mu_1(z) = \mathbb{E}[Y(1) \mid Z = z], \qquad \mu_0(z) = \mathbb{E}[Y(0) \mid Z = z],$$

describing how the means of the potential outcomes vary with covariates. The key step of the proof is

$$\underbrace{\mu_1(z)}_{\text{counterfactual}} = \underbrace{\mathbb{E}[Y \mid X = 1, Z = z]}_{\text{observational}}, \qquad \mu_0(z) = \mathbb{E}[Y \mid X = 0, Z = z].$$

5.5 Regression estimation of the ATE

Fit regressions $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$ (linear or non-linear); the safest practice is to estimate separate regressions in each treatment group. Then

$$\hat{\tau}_{\text{reg}} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)\right).$$

Procedure: predict for all units with $X_i = 1$; predict for all units with $X_i = 0$; average the differences.

5.6 Summary

Potential outcomes	$Y(1), Y(0)$
Conditional ignorability (back-door)	$Y(1), Y(0) \perp\!\!\!\perp X \mid Z$

Identification: $\mathbb{E}[Y(1)] = \mathbb{E}[\mathbb{E}[Y(1) \mid Z]] = \mathbb{E}[\mathbb{E}[Y(1) \mid X = 1, Z]] = \mathbb{E}[\mathbb{E}[Y \mid X = 1, Z]]$.

A Positivity and consistency in causal diagrams

Positivity concerns arrows into the treatment node. If X is a deterministic function of a pre-treatment Z (a bold/deterministic edge $Z \to X$), positivity is violated by design. In causal diagrams positivity is implicit unless otherwise specified.

Consistency concerns arrows out of the treatment node. $X \to Y$ must correspond to a well-defined intervention. If $X = x$ can be achieved in several ways with different effects on Y (weight loss via diet, exercise or surgery), the effect is ill-defined. Treatment nodes thus have a special status: they must correspond to sufficiently well-defined interventions (*What If*, §6.4).

B Confounding and exchangeability

- **Marginal exchangeability** $Y(x) \perp\!\!\!\perp X$ holds when X and Y have no common causes (no open back-door paths), as in a marginally randomised experiment ($X \to Y$ only).
- **Conditional exchangeability** $Y(x) \perp\!\!\!\perp X \mid Z$ holds when all back-door paths are blocked by Z, as in a conditionally randomised experiment ($Z \to X \to Y$, $Z \to Y$).
- If Z satisfies the back-door criterion, conditional exchangeability holds. The converse requires the more general adjustment criterion (Perković et al. 2018).
- Confounding (graphical, structural) = lack of exchangeability (counterfactual) (*What If*, §7.2 and Technical Point 7.1).

C Source coverage

The source deck has 60 physical pages (26 numbered slides with build-ups).

Topic	Physical pages
References, frameworks, PO overview	1–10
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SUTVA, interference, consistency, estimands	26–34
Unconfounded assignment, randomisation, SWIGs	35–44
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